MCT-SCINet: A Soil Moisture Prediction Model for Earthen Heritage Combining Mamba and Temporal Attention

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***Abstract*—Soil moisture is a critical parameter for the stability protection of earthen heritage, whose dynamic changes are influenced by complex multi-scale coupling and lag responses to key events. Existing models have significant limitations in long-term dependency modeling, multi-scale feature collaboration, and the ability to focus on critical time steps. To address these issues, this paper proposes the MCT-SCINet model, which achieves accurate prediction of soil moisture in earthen heritage by integrating Mamba state-space modeling, a hybrid convolutional-state structure, and a temporal attention mechanism. The core innovations include: 1) Introducing the Mamba module to replace traditional recursive structures, capturing long-term trends and cross-time dependencies through a dynamic parameter adjustment mechanism; 2) Designing a hierarchical hybrid encoder (ConvMixer and MambaMixer) to extract local capillary action features (hourly scale) and global environmental coupling features (monthly scale) in blocks; 3) Proposing a lightweight temporal attention mechanism to explicitly enhance the weight allocation of critical events such as rainfall and evaporation. Experiments show that on a multi-sensor dataset including different soil depths (5m, 10m, 15m), the model's MSE is 4.4% lower than SCINet (at 5m depth), with an R^2 of up to 0.85, and the error in predicting abrupt events is reduced by 18.6%. This study provides an efficient technical solution for real-time moisture regulation in earthen heritage protection.**

***Index Terms*—Earthen heritage protection; Soil moisture prediction; Improved SCINet; Mamba model; Temporal attention mechanism**

I. Introduction

China has a large number of earthen heritage sites with wide distribution, leading to heavy protection tasks and high demands. Soil moisture (SM) [1], [2] plays a crucial role in soil protection by maintaining soil structure, promoting plant growth, and influencing soil microbial activities. Meanwhile, soil moisture is one of the key factors affecting the stability of earthen heritage. Studies have shown that changes in soil moisture directly affect the mechanical properties of soil, such as shear strength, porosity, and saturation, thereby influencing the stability of earthen heritage. Therefore, accurate prediction of SM is of high value for the protection of earthen heritage. Currently, to improve the accuracy of surface SM prediction, researchers mainly focus on the Soil Moisture Active Passive (SMAP) product [2], which can effectively enhance the accuracy of surface SM prediction. The rapid development of deep learning has provided new solutions for accurate soil moisture prediction. Existing soil moisture prediction methods can be mainly divided into time-series modeling-based methods, spatio-temporal joint modeling-based methods, and multi-modal data fusion methods.

Time-series modeling-based methods, such as traditional time-series models like Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) [3], [4], capture temporal dependencies through recursive structures and perform stably in single-point moisture prediction. Transformer models break through the sequence length limitation via self-attention mechanisms [5], but their quadratic computational complexity restricts their application in dense sensor networks. Spatio-temporal joint modeling-based methods model the spatial correlation of sensors through graph structures. STGCN [6] aggregates neighborhood information using predefined adjacency matrices, but its static graph construction strategy cannot adapt to the dynamic changes in soil permeability and microtopography. Multi-modal data fusion methods for soil moisture prediction require synergizing meteorological data (temperature, precipitation), soil properties (porosity, salinity), and multi-sensor time-series signals. Despite progress, existing methods face the following limitations in multi-sensor scenarios for earthen heritage:

A. Difficulty in capturing global dependencies in CNN/ SCINet-like models:

Although many models based on convolutional [7], [8] structures (e.g., SCINet, [9]TCN) can effectively extract localized patterns, such as short-period humidity fluctuations, they have limited receptive fields and are difficult to capture trend information over longer time domain scales, such as persistent droughts and processes such as the rebound of soil water content. This local modeling approach is prone to ignore

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long-distance information transfer, which leads to trend bias and affects the stability of long-term prediction.

B. Single-model structure failing to balance dynamic modeling across different time scales:

Soil moisture series usually contain both short-period (e.g. evaporation fluctuations due to diurnal alternation) and long-period (e.g. monthly climate change) components. It is often difficult for single-structure models to accommodate both types of feature. For example, a pure convolutional model can only extract patterns within a fixed sensory field, while a pure Transformer model, although modeling a wide range, is unstable and prone to overfitting under small samples or high-frequency noisy data.

C. Lack of explicit modeling mechanisms for critical time steps:

In soil moisture prediction scenarios, certain key events (e.g., irrigation initiation, heavy rainfall, etc.) can lead to drastic changes in soil moisture, which in turn affects the dynamics at multiple subsequent time steps. However, most of the structures based on convolutional or state modeling lack a focusing mechanism for specific moments in the time dimension, i.e., the model treats each time step equally, and it is difficult to explicitly identify and enhance the representation of these key influencing points, which leads to an increase in the prediction error of the model near the abrupt change point. Based on the above problems, this paper proposes a hybrid neural network structure that integrates state space modeling [10] and temporal attention mechanism —— MCT-SCINet, which aims to solve the contradiction between long-term dependent modeling, critical time step attention and computational efficiency of existing models. Specifically, this paper innovatively presents: 1.The first introduction of the Mamba model [10]–[12] into soil moisture prediction models to replace partial convolutional structures, enhancing the model's ability to model long-term dependencies.
2. Adoption of a hybrid convolutional-Mamba structure to retain the efficiency of CNN while introducing Mamba's global state modeling capability, constructing a more balanced time modeling framework.
3. Introduction of a lightweight temporal attention mechanism [13], [14]after the hybrid structure to guide the model to focus on critical moments by assigning weights to different time steps, thereby improving overall prediction performance and local sensitivity.

II. Related Work

A. Time Series Prediction Models

Time series prediction is a cornerstone in fields like meteorology, transportation, and energy. Recurrent models such as LSTM and GRU [3]tackle long-term dependencies but grapple with gradient vanishing/exploding issues. Transformer's self-attention revolutionized sequence modeling but at the cost of quadratic computational complexity. SCINet [9] improves efficiency through hierarchical convolution yet its fixed receptive fields hinder adaptability to multi-scale dynamics. The Mamba model, based on State Space Models (SSM), introduces linear-complexity dynamic receptive fields, though its application in multi-sensor spatio-temporal data remains largely unexplored.

B. Multi-Modal Data Fusion Methods

Early fusion techniques (e.g., feature concatenation) oversimplified interactions between heterogeneous data sources. Attention mechanisms (e.g., STGCN) enable dynamic weighting of modalities but struggle to suppress redundant or conflicting information. Gated mechanisms, effective in speech and image tasks, have yet to be fully integrated into time-series multi-modal fusion frameworks.

C. Applications in Soil Moisture Prediction

Physics-based models (e.g., HYDRUS) demand precise parameterization and high computational resources. Data-driven approaches (LSTM/CNN) overlook spatial correlations in multi-sensor networks. Spatio-temporal graph models (STGCN/Transformer) rely on static graph structures, failing to account for dynamic soil properties. Existing studies in earthen heritage protection lack comprehensive strategies to address spatio-temporal heterogeneity and noise interference across sensors, limiting predictive accuracy and robustness.

III. Methodology

To enhance the modeling capability for complex soil moisture time series, this paper proposes MCT-SCINet, a hybrid neural network model integrating state-space modeling, deep structure mixing, and temporal attention mechanisms. Built upon the original SCINet architecture, the model achieves efficient modeling of local disturbances, long-term trends, and critical time nodes through three key innovations:

- Introducing the state-space model Mamba during prediction to enhance long-term dependency modeling.
- Designing a hierarchical hybrid structure of ConvMixer and MambaMixer to effectively combine local and global modeling.
- Introducing a temporal attention fusion module before prediction output to explicitly model the importance of critical time steps.

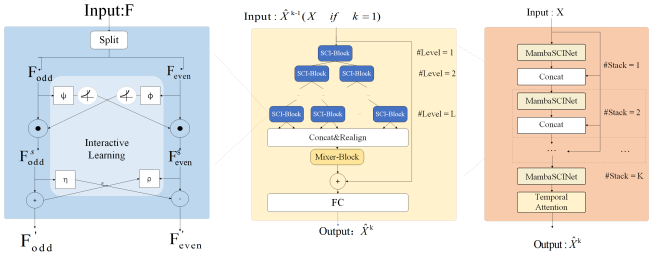


Fig. 1. System Architecture Diagram

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A. Introduction of the Mamba Module: Enhancing State Dependency Modeling

To overcome the limited receptive field of traditional convolutional models, this paper introduces the state-space modeling module Mamba [11] into the model (Fig. 2). This module effectively models long-range temporal relationships, particularly suitable for trend changes, periodic transitions, and the propagation of remote influencing factors in soil moisture sequences.

The continuous-time SSM can be expressed as:

$$\begin{cases} \mathrm{x}'(t) = \mathrm{A}\mathrm{x}(t) + \mathrm{B}\mathrm{u}(t) \\ \mathrm{y}(t) = \mathrm{C}\mathrm{x}(t) + \mathrm{D}\mathrm{u}(t) \end{cases} \tag{1}$$

where $\mathbf{x}(t) \in \mathbb{R}^N$ is the hidden state vector, $\mathbf{u}(t) \in \mathbb{R}$ is the input signal, and $\mathbf{y}(t) \in \mathbb{R}^N$ is the output signal. $\mathbf{A} \in \mathbb{R}^{N\times N}$ controls the state evolution, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, $\mathbf{C} \in \mathbb{R}^{1\times N}$, and $\mathbf{D} \in \mathbb{R}$ are parameter matrices.

Discretization and Dynamic Parameter Adjustment: Mamba converts the continuous system into a discrete form using the Zero-Order Hold (ZOH) discretization method and introduces learnable parameters to dynamically adjust A:

$$\bar{\mathbf{A}} = \exp(\Delta \cdot \mathbf{A}), \quad \bar{\mathbf{B}} = (\Delta \cdot \mathbf{A})^{-1}\big(\exp(\Delta \cdot \mathbf{A}) - \mathbf{I}\big)\mathbf{B} \tag{2}$$

$$\mathbf{x}(t) = \bar{\mathbf{A}}\mathbf{x}(t) + \bar{\mathbf{B}}\mathbf{u}(t) \tag{3}$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \tag{4}$$

where Δ is a learnable step-size parameter that controls the time granularity of state transitions. The matrix $\mathbf{A}$ is computed efficiently through structured parameterization to avoid the high computational complexity of traditional SSMs.

As the core computational module of MambaMixer, Mamba combines input projection and output mapping to update and propagate state information. In the decoder, Mamba modules are directly used to construct deep state evolution structures, stacked consecutively to enhance the model's memory and abstraction capabilities in the time dimension. The introduction of Mamba gives the model a more efficient and stable long-term modeling capability than traditional convolutional or attention structures, especially suitable for agricultural environment scenarios requiring long historical information for prediction.

B. MixerBlock: Deep Hybrid Structure

Given that soil moisture sequences exhibit both short-term disturbances and long-term trends, single-structure models often fail to balance modeling across different time scales. Soil moisture changes are driven by long-term climatic factors (e.g., seasonal rainfall, evaporation), while traditional Convolutional Neural Networks (CNNs) have limited receptive fields and cannot model cross-month or even cross-year dependencies. Mamba, based on the state-space model (SSM), explicitly captures long-term trend structures by dynamically adjusting state transmission parameters. This paper proposes a hybrid convolutional-state stacking approach, where the first few layers use ConvMixer to capture local information, and the subsequent layers use MambaMixer to model global dynamics (Fig. 2):

$$\text{MixerBlock} = \underbrace{[\text{ConvMixer}_1, \ldots, \text{ConvMixer}_k]}_{\text{Local Feature Extraction}}, \underbrace{[\text{MambaMixer}_{k+1}, \ldots, \text{MambaMixer}_L]}_{\text{Long-term Modeling}} \tag{5}$$

This structure combines the advantages of both model types: the MixerBlock is composed of multiple stacked modules to extract multi-scale temporal features of the input sequence, balancing the modeling of short-term local patterns and long-term global trends. The first few layers use ConvMixer to extract local features, while the subsequent layers use MambaMixer to capture cross-time-step dependencies.

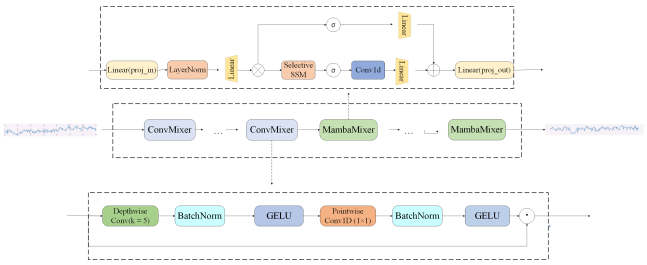


Fig. 2. MixerBlock

ConvMixer provides efficient local pattern modeling: It accurately captures humidity abrupt changes within 0-24 hours after irrigation/rainfall. ConvMixer's 5-hour window can locate moisture penetration peaks, avoiding the detail blurring caused by traditional models with overly large windows (e.g., LSTM averaging a humidity surge over 12 hours into a gentle change)

$$\mathbf{X} = \text{GELU}(\text{BN}(\text{DConv1d}(5, \mathbf{X}))) \odot \text{PConv1d}(1, \mathbf{X}) \tag{6}$$

MambaMixer expands the receptive field and models cross-time dependencies: The module encapsulates a Linear $\rightarrow$ Mamba $\rightarrow$ Linear structure, enabling deeper modeling of global state propagation characteristics.

$$\mathbf{X} = \mathbf{X} + \text{Mamba}(\text{LayerNorm}(\mathbf{x})) \tag{7}$$

C. Temporal Attention Fusion Module

After the MixerBlock output, the model introduces a lightweight temporal attention module [13], [14] to weight the importance of different time steps. Taking the decoder output sequence as input, this module calculates attention weights between each time step and reconstructs information aggregation accordingly.

This mechanism adaptively identifies critical moments in the sequence, such as sudden rainfall or irrigation points, guiding the model to focus more on these high-impact areas during prediction. Compared with traditional uniform weighting, the temporal attention mechanism significantly improves prediction accuracy near abrupt change points.

The MCT-SCINet model enhances prediction capabilities for complex time series like soil moisture through three structural innovations:

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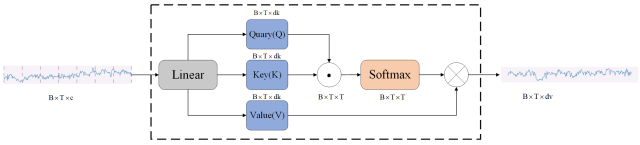


Fig. 3. Temporal Attention

- Deep hybrid structure of convolution and Mamba, achieving effective collaboration between local and global modeling.
- Temporal attention fusion mechanism, improving the model's response to critical time steps (e.g., abrupt change points).
- First introduction of the Mamba module in soil moisture prediction models, enhancing the ability to model long-term trends and remote temporal relationships.

IV. Experiments and Analysis

A. Dataset Description

This study uses a self-constructed soil moisture dataset for earthen heritage, To further improve the model performance, we conducted Pearson correlation coefficient analysis on the correlation between each environmental indicator and soil moisture, and found that the amount of change, evapotranspiration, and temperature were highly correlated with moisture; whereas the correlation of indicators such as wind speed was weak. The indicators with significant correlations were finally retained as input features.Collected from a typical earthen heritage area with the following characteristics:

Sensor layout: Three soil moisture sensors deployed at depths of 5m, 10m, and 15m below the surface to capture humidity dynamics across different soil layers.

Time range: January 1, 2021, to December 31, 2023, with hourly synchronous data collection, totaling 26,280 time-series samples per sensor.

Data division:

- Training set: January 2021 – June 2022 (13,140 samples)
- Validation set: July 2022 – December 2022 (3,942 samples)
- Test set: January 2023 – December 2023 (5,256 samples)

Preprocessing:

- **Sliding window method** to generate input-output sequences: input window length 96 hours (historical humidity and environmental features), prediction window length 48 hours (target humidity).
- **Z-score standardization** for numerical features:

$$x' = \frac{x - \mu}{\sigma} \tag{8}$$

where μ, σ are the mean and standard deviation of the training set.

- For multi-sensor data, features from the three depths are concatenated by dimension into an input matrix $\mathbf{X} \in \mathbb{R}^{T\times(3+8)}$, where the first 3 dimensions are humidity at each depth, and the last 8 dimensions are environmental variables.

B. Experimental Setup

Comparison models:

1) Traditional time-series models: LSTM [7], GRU (hidden layer dimension 64, 2 layers)
2) Convolution-based models: SCINet (original architecture, kernel size 3), TCN (dilation rates $[1, 2, 4]$)
3) Attention-based models: Informer [5] (single head, hidden layer dimension 4)
4) Spatio-temporal models: STGCN (graph structure built based on sensor physical distances)

Evaluation metrics:

TABLE I
Common Regression Evaluation Metrics and Their Meanings

Metric	Formula	Physical Meaning
MSE	$\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2$	Overall prediction bias, sensitive to outliers
MAE	$\frac{1}{N}\sum_{i=1}^{N}\lvert\hat{y}_i - y_i\rvert$	Linear error metric, robust to outliers
RMSE	$\sqrt{\text{MSE}}$	Standard deviation of prediction errors
R^2 Score	$1 - \frac{\sum(\hat{y}_i - y_i)^2}{\sum(y_i - \bar{y})^2}$	Goodness of fit between prediction and ground truth

Training parameters: Optimizer: AdamW (weight decay 1e-4); Learning rate: Initial 1e-3, cosine annealing decay; Batch size: 32; Epochs: 100; Hardware: NVIDIA RTX 3090 GPU, PyTorch 2.5 framework.

C. Comparative Experiments with Other Algorithms

To validate the superiority of the proposed algorithm, comparative experiments were conducted with other time-series prediction models under the same training parameters. (Due to space constraints, results for the 5m-depth sensor are presented as an example.)

TABLE II
Comparison Experiments

Depth	Model	MAE	MSE	RMSE	R^2
5 m	Transformer	0.1521	0.1127	0.3357	0.81
	Informer	0.1477	0.0788	0.2807	0.84
	GRU	0.1460	0.1021	0.3195	0.84
	STGCN	0.1390	0.0954	0.3088	0.83
	mamba	0.1422	0.1103	0.3321	0.84
	SCINet	0.1388	0.0766	0.2767	0.84
	MCT-SCINet	**0.1321**	**0.0732**	**0.2705**	**0.85**

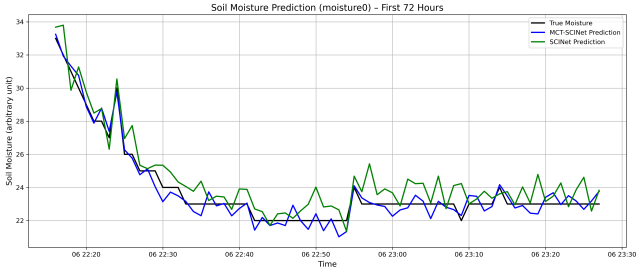


Fig. 4. Prediction Curve Plot(One-hour slice at a depth of 5m)

In the 5m-depth dataset, MCT-SCINet achieved a MAE of 0.1321, a 4.8% reduction compared to the baseline SCINet

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(0.1388), a 4.4% reduction in MSE (0.0732), and an R^2 improvement to 0.85. These results indicate:

- Effectiveness of multi-scale feature fusion: The hybrid architecture captures local disturbances (e.g., day-night evaporation fluctuations) via ConvMixer and models long-term trends (e.g., monthly rainfall accumulation) via MambaMixer, significantly enhancing the model's representation of complex temporal patterns.
- Optimization of dynamic parameter adjustment: The Mamba module adaptively adjusts state transmission granularity through input-dependent step-size parameters and structured parameterization (diagonal matrix A), overcoming the fixed receptive field limitation of traditional convolutional models.

Comprehensive evaluations were conducted on the self-constructed multi-depth soil moisture dataset (5m, 10m, 15m) by comparing MCT-SCINet with classic time-series prediction models (LSTM, GRU, SCINet, Informer, STGCN) to validate its performance advantages.

D. Ablation Experiments

TABLE III
ABLATION EXPERIMENTS

Model	MAE	MSE	RMSE	R^2
SCINet	0.1421	0.0766	0.2767	0.84
SCINet+C	0.1537	0.1012	0.3181	0.82
SCINet+T	0.1443	0.1289	0.3590	0.83
SCINet+M	0.1411	0.0935	0.3057	0.84
SCINet+M+T	0.1424	0.0832	0.2884	0.85
SCINet+M+C	0.1412	0.0972	0.3117	0.83
SCINet+T+C	0.1454	0.0832	0.2884	0.84
SCINet+M+T+C	**0.1321**	**0.0732**	**0.2705**	**0.85**

To validate the effectiveness of each improvement strategy, we conducted ablation studies. Here, C denotes ConvMixer, M denotes MambaMixer, and T denotes Temporal Attention. Introducing the MambaMixer module alone (SCINet+M, MAE=0.1411) or temporal attention (SCINet+T, MAE=0.1443) showed limited effects, while the synergistic combination of all three modules (SCINet+M+T+C, MAE=0.1321) achieved optimal performance. This indicates that:

- Complementarity of modules: Mamba handles long-term dependency modeling, ConvMixer extracts local features, and temporal attention focuses on critical events—all three are indispensable.
- Relying on a single module (e.g., pure Mamba or pure attention) is prone to noise interference, whereas the hybrid structure enhances noise resistance through stage-wise feature extraction.

V. CONCLUSION

This paper addresses the challenges of multi-scale dynamic modeling and critical event response in soil moisture prediction for earthen heritage by proposing the MCT-SCINet model, which integrates Mamba state-space modeling and temporal attention mechanisms. By introducing the Mamba module to enhance long-term trend capture, combining a hybrid convolutional-state architecture to synergistically extract local disturbances and global features, and using temporal attention to focus on abrupt events, the model excels in complex time-series prediction tasks. Experiments show that compared with baseline models like LSTM and SCINet, MCT-SCINet achieves optimal performance in MAE, MSE, and R^2, particularly reducing errors in predicting abrupt events (e.g., heavy rainfall). The research provides theoretical support and technical tools for stability monitoring and precise moisture regulation in earthen heritage. Future work will further integrate multi-source heterogeneous data (e.g., vegetation cover, microbial activity) and optimize the model's robustness in extreme climate scenarios to promote its practical application in cultural heritage protection.

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